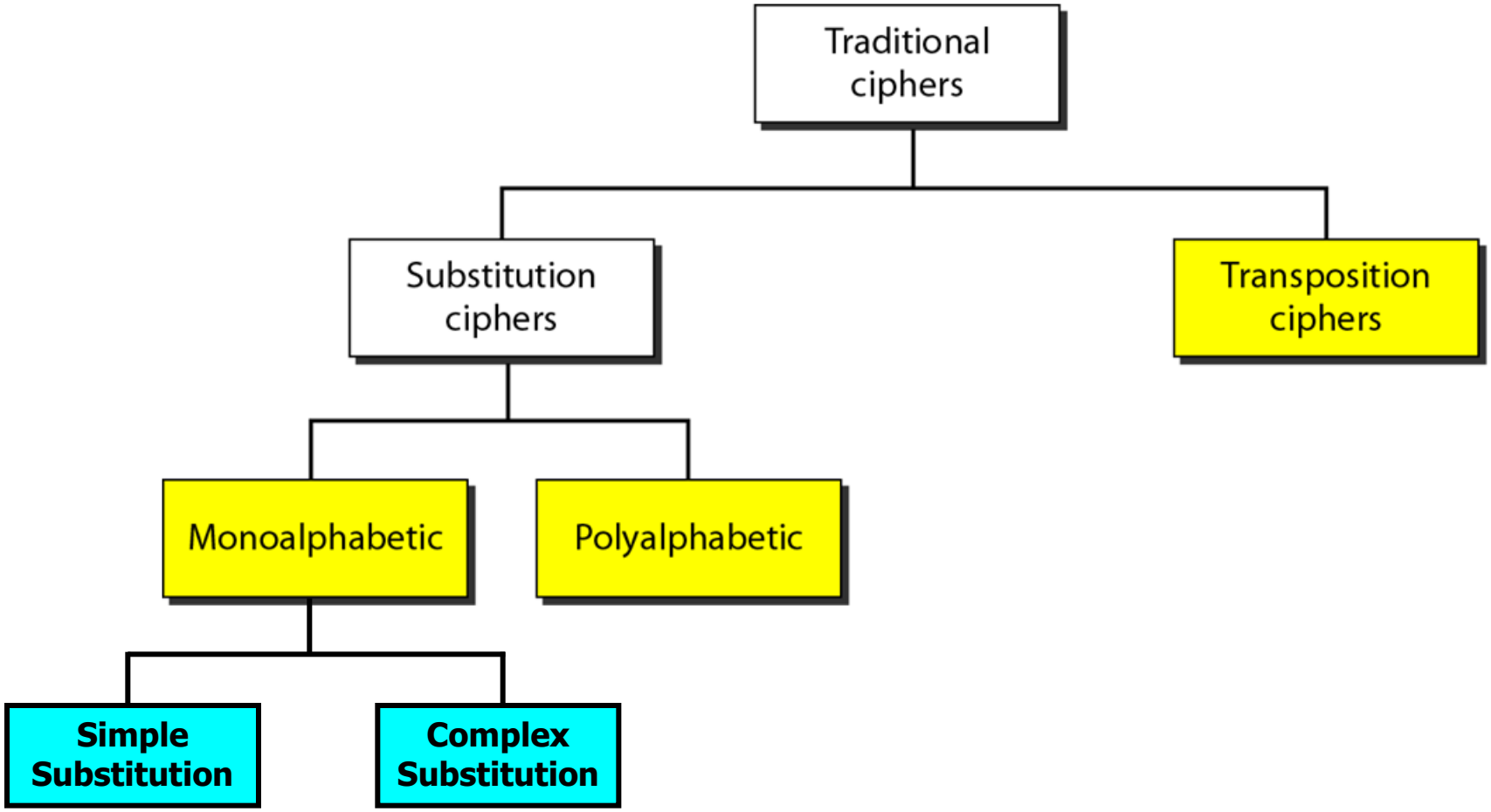
Substitution Ciphers

Monoalphabetic cipher



Substitution Ciphers

Monoalphabetic cipher

Simple Substitution

- Replaces each letter in the plaintext by a letter some fixed number of positions further down the alphabet
- It is also called **Shift cipher**
- It is also called **Caesar cipher.**

We studied this cipher in the previous lecture.

Implementation

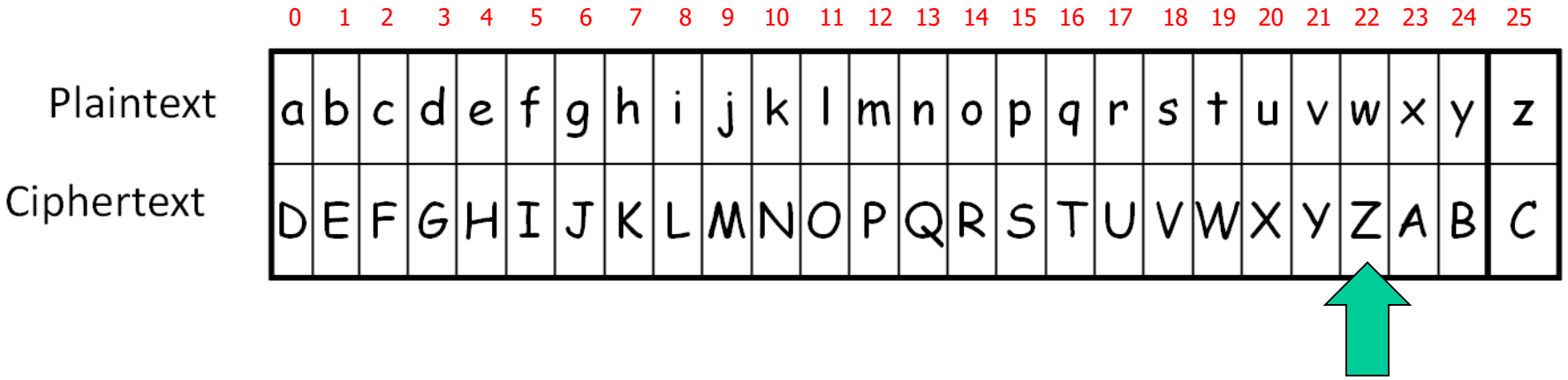
$$C = E(p) = (p + k) \bmod (26)$$

$$p = D(C) = [(C - k) \mathbf{+26}] \bmod (26)$$

Here:
p = plaintext
C= Ciphertext
k= key

Possible Remainders: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

Shift by 3



- **Example:**

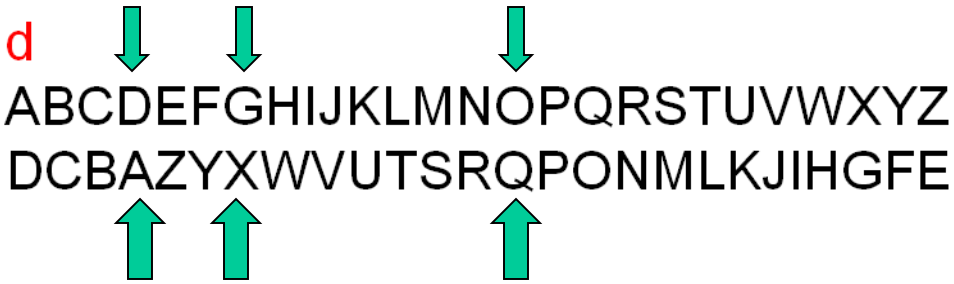
Plaintext: we love pakistan

Ciphertext: ZH ORYH SDNLVWDQ

- Advantages:
  - Simple
  - Fast
- Disadvantages:
  - Easy to break

- **EXAMPLE:**
  - What is **ciphertext** when **key** is **d**(**3**) and **plaintext** is **GOOD**

Key:



Plaintext: GOOD

Ciphertext: XQQA

- **EXAMPLE 2:**
  - What is **ciphertext** when **key** is **f**(5) and **plaintext** is **GREAT**

KEY	f																							
Plaintext	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Ciphertext																										

Plaintext: GREAT

Ciphertext:

Example:

- Keyword: deceptive
- key: deceptivedeceptivedeceptive
- plaintext: wearediscoveredsaveyourself
- ciphertext: ZICVTWQNGRZGVTWAVZHCQYGLMGJ

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Example:

- Keyword: deceptive

- key: deceptivewearediscoveredsav
- plaintext: wearediscoveredsaveyourself
- ciphertext: ZICVTWQNGKZEIIGASXSTSLVVWLA

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Transposition Ciphers

Row Transposition Ciphers

a t t a c k p o s t p o n e d u n t i l t w o a m

Known=key

Key: 3 4 2 1 5 6 7

Plaintext:

a t t a c k p
o s t p o n e
d u n t i l t
w o a m **xyz**

Filling empty spaces from end towards start. i.e. z then y then x then w, ……

Ciphertext: TTNAAPTMTSUOAODWCOIXKNLYPETZ

3 4 2 1 5 6 7

Transposition Ciphers

Row Transposition Ciphers

CLASS EXERCISE

MESSAGE: THIS IS VERY HARDWORKING CLASS

Key: 3 4 6 5 2 1

What is the ciphertext?

Transposition Ciphers

Double Transposition

- Plaintext: attackxatxdawn

	1	2	3
1	a	t	t
2	a	c	k
3	x	a	t
4	x	d	a
5	w	n	x

Permute Rows

	1	2	3
3	x	a	t
5	w	n	x
1	a	t	t
4	x	d	a
2	a	c	k

Permute Columns

	1	3	2
3	x	t	a
5	w	x	n
1	a	t	t
4	x	a	d
2	a	k	c

- Ciphertext: xtawxnattxadakc
- Key is matrix size and permutations: (3,5,1,4,2) and (1, 3, 2)

ROWS

COLUMNS

Transposition Ciphers

Reverse Cipher

- Write the message backwards

Example:

- Plain: I CAME I SAW I CONQUERED
- Cipher: DEREU QNOCI WASIE MACI

Transposition Ciphers

Rail Fence Cipher

Write message with letters on alternate rows
Read off cipher row by row

Example:

- Plain: I A E S W C N U R D
  C M I A I O Q E E
- Cipher: IAESW CNURD CMIAI OQEE

Transposition Ciphers

Rail Fence Cipher

Example:

Plaintext: m e m a t r h t g p r y

e t e f e t e o a a t

Cipher: MEMATRHTGPRYETEFETEOAAT

How to find Plaintext from Ciphertext?

- Count characters from Ciphertext.
- Divide number of characters by 2.
- Break the Ciphertext into two portions according to above division.
- Write in two lines.
- Give space between characters.
- Combine two lines so that 1st character from 1st line, then 1st character from 2nd line and so on.

Transposition Ciphers

Rail Fence Cipher

Example:

If we have 3 rails and a message of "This is a secret message", you would write out:

```
T S A C T S G
H I S R M S E
I S E E E A J
```

The last J is just a random letter to fill in the space.

The secret message is then condensed and regrouped.

TSACT SGHIS RMSEI SEEEA JGURL

Transposition Ciphers

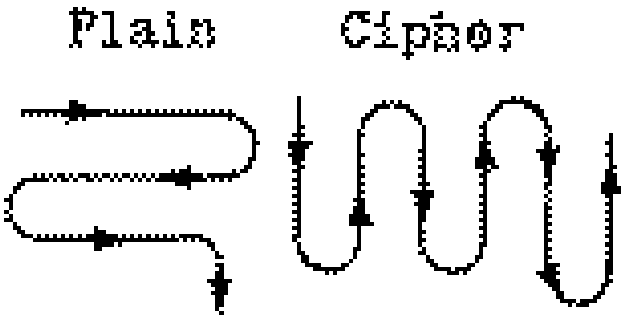
Geometric Figure

Write message following one pattern and read out with another

Example:

Given = No of columns = 6

I	C	A	M	E	I
O	C	I	W	A	S
N	Q	U	E	R	E
					D



Cipher: IONQC CAIUE WMEAR DESI

How to find Plaintext from Ciphertext?

Playfair Cipher

EXAMPLE :

- Keyword : PAPPU
- Message : WRAPPINGOOP
- Filling Character: Q

- Encryption:
  - Pairs:
  - WR AP PI NG **OQ** OP
  - XQ UA DO MH QR VD

PAPPU				
P	A	U	B	C
D	E	F	G	H
I/J	K	L	M	N
O	Q	R	S	T
V	W	X	Y	Z

Playfair Cipher

EXAMPLE :

- Keyword : ABBAAMMI
- Message : ALLFAMILYWENTTOZOO
- Filling Character: Q

- Encryption:
  - Pairs:
  - AL LF AM IL YW EN TQ TO ZO OQ
  - BK NE BI BO ZX FL UR YT YP KT

ABBAAMMI
A B M I/J C
D E F G H
K L N O P
Q R S T U
V W X Y Z

Playfair Cipher

EXAMPLE :

- Σ = {A B C D E F G H I J K L M N O P Q R S T U V W X Y Z}
- Keyword : ABBAAMMI
- Message : PAPPUWENTTOZOO
- Filling Character: Q
- Encrypt it

Solve it.

ABBAAMMI
A B M I/J C
D E F G H
K L N O P
Q R S T U
V W X Y Z

Playfair Cipher

EXAMPLE :

- Σ = {A B C D E F G H I J K L M N O P Q R S T U V W X Y Z 0 1 2 3 4 5 6 7 8 9}
- Keyword : ABBAAMMI4US
- Message : PAPPUWENT2ZOO4LIONS
- Filling Character: Q

- Encrypt it

Solve it.